

→ Leetcode 9912

constraints:

- $1 \leq \text{nums length} \leq 5 \times 10^4$
- $-5 \times 10^4 \leq \text{nums}[i] \leq 5 \times 10^4$

```
void merge(int* nums, int left, int mid, int right) {
    int i = left, j = mid+1, k = right;
    int* temp = (int*) malloc((right-left+1)*sizeof(int));
    while (i <= mid && j <= right) {
        if (nums[i] <= nums[j])
            temp[k++] = nums[i++];
        else
            temp[k++] = nums[j++];
    }
    while (i <= mid)
        temp[k++] = nums[i++];
    while (j <= right)
        temp[k++] = nums[j++];
    for (i = left, k = 0; i <= right; i++, k++) {
        nums[i] = temp[k];
    }
    free(temp);
}
```

```
void mergesort(int* nums, int left, int right) {
    if (left >= right) return;
    int mid = (left + right) / 2;
    mergesort(nums, left, mid);
    mergesort(nums, mid+1, right);
    merge(nums, left, mid, right);
}
```

```
int** sortArray(int* nums, int numSize, int* returnSize) {
    mergesort(nums, 0, numSize-1);
    *returnSize = numSize;
    return nums;
}
```

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